

Detecting unknown arbuscular mycorrhizal fungi from linked long-read rDNA copies

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Abstract

1. Arbuscular mycorrhizal fungi (AMF) are named in environmental DNA by three systems that rarely agree: MaarjAM virtual taxa (VT) from the SSU, UNITE species hypotheses (SH) from the ITS, and clades on LSU reference trees. We asked how reliably a call combining the three names known species and flags unknown ones.
2. We cut 903 long-read rDNA copies from 151 cultures of 37 named species into the barcode windows each database was built on, linked the three systems on the same molecules, and combined compatible units into one call, with every link fitted on training cultures only. We withheld species from the cultures and the LSU reference, and tested 52 cultures from other collections (LongGloDB) against locked predictions.
3. The systems mostly nest, SHs inside LSU species inside VTs, but VTs resolve at most 19 of 37 species, and the main-type copies of 74 of 107 cultures span 2–6 SHs. Copies with a shortened 5.8S mimic new lineages. Main-type copies of species held by another culture kept a compatible known call in 745 of 755 cases. For withheld species, 11.6% of copies were called known, and 7.9% after a culture-ITS distance rule; with the species left in the LSU reference these fell to 2.8% and 1.3%, so much of the novelty signal came from the external reference. The genus of a withheld species was kept for only 66.9% of copies whose genus another culture holds. All 53 remaining errors lie inside 4 species complexes (9 species) whose members overlap on all three markers; one exists only at the 99th percentile used to define them. With MaarjAM and UNITE intact, this may understate errors for species without a VT by up to 2.7% of copies. On LongGloDB, 4 of 102 ASVs of species absent from the reference were called known, all as predicted; there the ITS rule prevented no error and cost 15 of 164 known-species calls.
4. Novelty can be called reliably only above the resolution of the markers, and species they cannot separate should be reported as groups. Linked long-read copies also bridge the three naming systems empirically.

Keywords. arbuscular mycorrhizal fungi; Glomeromycota; rDNA intragenomic variation; MaarjAM; UNITE; novelty detection; phylogenetic placement; long-read metabarcoding.

1 Introduction

AMF surveys name what they find in three incompatible ways, so results from different markers cannot be pooled. SSU studies report MaarjAM virtual taxa [27, 26]; ITS studies report UNITE species hypotheses [17, 23]; LSU studies place reads on curated reference trees [19, 9, 7, 8]. No agreed table maps one system onto another, and global occurrence databases keep the datasets apart [35]. The names in the three databases were assigned independently, many reference sequences are unidentified environmental clones, and the taxonomy itself keeps moving: the family Glomeraceae alone has recently been divided into several families [6].

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AMF genomes make the mapping harder. Their rDNA copies are not homogenised: copies within one individual differ [28], a single *Rhizophagus irregularis* genome carries only ten or eleven complete 45S units that differ in sequence and are not arranged in tandem [22], and copies within a culture can differ by several percent in ITS [34, 32]. One genome can therefore look like several taxa in ITS, and a divergent copy can look like an unknown lineage. Any claim of AMF novelty has to rule these out first. Novelty calls have been calibrated for fungal ITS generally [25], but not for the AMF case, where three naming systems and intragenomic variation interact.

Long reads spanning the SSU, ITS and LSU link all three markers on one molecule. Amplicons of about 2.5 kb were introduced to consolidate the three barcoding regions into one dataset [16], and a large collection of such copies from identified cultures now exists [32]. It lets each copy be named by all three systems at once, and each culture serves as ground truth for its own copies. We use it to ask four questions:

1. How do VTs, SHs and LSU clades map onto each other and onto cultured species?
2. Which divergent copy classes exist, and how does each mislead naming?
3. Can compatible calls from the three markers be combined into one reliable name?
4. How often is a species the reference does not know called a known species, and how much does the answer depend on what the external databases contain?

This work comes from amfsplit, a long-read AMF metabarcoding tool built on ITSxRust v0.3.0 [24]. We report the part that bridges the naming systems and tests novelty calls (Stage 3). Cutting reads into the legacy windows (Stage 1) is described only as far as the deposited copies need; its performance on environmental libraries, and the copy-aware abundance model (Stage 2), will be reported with the tool.

2 Methods

Every step below reads a frozen, versioned culture reference, and every quantity learned from cultures is refitted inside each validation fold.

2.1 Regions and legacy windows

Anchors and ITS regions. ITSxRust anchors each read with one nhmmer pass (HMMER 3 [11], ITSx fungal profile [3]) and delimits the SSU flank, ITS1, 5.8S, ITS2, full ITS and LSU flank. We use tolerant anchoring thresholds (E-value $\leq 10^{-3}$, anchor score ≥ 15): anchoring only delimits regions.

Legacy windows. A panel of seven windows (Table 2) is cut at the original primer sites, so each window covers the region its legacy database was built on. The panel uses the primers of the SSU virtual taxa (NS31, AM1, AML2 [31, 14, 21, 27]), the SSU V4 primers WANDA and AMV4.5NF/AMDGR [10, 29], the LSU primers LROR [4, 9] and FLR2 [33], the D2 primers FLR3 and FLR4 [13], and the outer pair AML1 and LSUmBr [21, 20] of the long amplicon [16], LSUmBr as the degenerate consensus of its five variants [30]. Primers are located with an IUPAC-aware infix alignment (edlib [36]) that tolerates edits up to 15% of the primer length; windows are primer-trimmed, and coordinates are 1-based, inclusive and in rRNA-sense orientation. Every primer sequence was checked against published sources (`analysis/primer_check.tsv`). A primer is searched only in the part of the read where it can lie (SSU primers before the ITS, LSU primers after it, plus 50 nt). A window is *open* when a primer site lies beyond the end of the copy, and *missing* when the primer is absent although the copy covers its expected position.

Copies were also screened for repeats (concatemers) with the screen of amfsplit version 0.3.1, which will be described with the tool.

2.2 Data and regions

We used the 904 rDNA copies deposited by Stefani et al. [32] (GenBank PX207096–PX207274 and PX214440–PX215164). Their study reports PacBio rDNA for 148 cultures of 44 species, and the three protein-coding genes for 143 cultures of 39 phylogenetic species; only the rDNA copies are used here. The deposit carries 148 culture labels and, in its organism fields, 41 names (36 species and 5 named only to genus), the same names as their culture tables (Tables S2 and S3 there). Their 39 protein-coding species are phylogenetic species, which split *Septoglomus* sp. into three; the 44 rDNA species are not itemised, and we use the deposit’s names. Three labels hold two lineages each (mixed cultures), which we count as separate cultures, hence 151. After the name edits (Table 3) the reference holds 41 names: 37 named species and 4 named only to genus. Stage 1 cut each copy into the SSU flank, ITS1, 5.8S, ITS2, full ITS, the LSU LROR–FLR2 window and the D2 window; all 904 copies were anchored. Sequences were compared with edlib [36] using infix edit distance as a percentage of the shorter sequence.

2.3 Culture reference

A culture is an organism name plus a culture label. `freeze_ref.py` builds the reference from the Stage 1 tables and an edits file. Every edit states first what the data must look like (organism, label, copy count, VT, copy class, 5.8S length), and a failed expectation stops the build before anything is written. Each version records input and output hashes and a fingerprint. The edits leading to version 4 (fingerprint `181938d695f3df40`) are listed in Table 3.

Copy classes follow the 5.8S. Stage 1 flagged copies whose 5.8S is more than 3% from the culture medoid. We split the flagged copies into *short-5.8S* copies, at least 3 bp shorter than the culture’s main type, and *same-length* copies. Where the medoid fell inside the short-5.8S group the flags came out inverted, and the classes were swapped (Table 3).

2.4 External references and assignment arms

The VT arm used the MaarjAM gDAT snapshot of July 2020 [27], the SH arm UNITE release 10.0 of 19 February 2025 [23, 1], and the LSU arm the Delavaux V18 database of May 2025 [9, 7, 8] (repository commit `f4a0014`). Searches used BLAST+ [5] with the settings in Table 1. Many V18 records carry terminal N padding, so coverage does not count reference Ns; without that correction 95 of 460 records fail the coverage test even against themselves. Hits to any accession of the culture reference were dropped, so no copy could take a name from itself or a culture-mate. A culture’s label on an arm is the modal label of its main-type copies (ties go to the higher median identity). A copy’s calls depend only on its window, the database and the dropped accessions, not on culture names. The crosswalk was run before the last two *Septoglomus* edits (Table 3), so its culture-level report was replayed on the final labels with the stored per-copy calls (`crosswalk_replay.py`), after checking every stored call against per-unit evidence captured on the final reference: on all three arms and all 903 copies the call’s identity equals that of the highest-scoring hit passing the threshold, and the best unit is present at that identity.

2.5 Compatible unit sets

Each copy receives a set of units instead of a single best hit. The anchor is the copy’s best hit by bit score, and the set holds every unit within δ identity points of the anchor that also passes the identity threshold. Units that split one culture’s main-type copies at $\delta = 0$ are merged when the closest copies across the split are within 1% in the SSU flank; that limit keeps the two lineages

Table 1: The three assignment arms. Each copy is searched with BLAST+ against one external reference per arm; all arms require e-value $\leq 10^{-50}$.

Arm	Query region	Task	Reference	Identity	Coverage
VT	SSU flank	megablast	MaarjAM (gDAT, July 2020; 3,081 seq., 384 VT)	$\geq 97\%$	$\geq 95\%$
SH	full ITS	blastn	UNITE 10.0, UNITE+INSD (16,800 Glomeromycota seq., 7,563 SH)	$\geq 98.5\%$	$\geq 90\%$
LSU	LROR-FLR2	blastn	Delavaux V18 (460 AMF records, 234 species labels)	$\geq 97\%$	$\geq 90\%$

Coverage is of the shorter sequence, with reference Ns not counted. Hits to any accession of the culture reference are dropped. The SH arm was also run against the UNITE general release (2,048 Glomeromycota representatives).

of a mixed culture from being merged. δ is the smallest margin at which the main-type copies of at least 99% of cultures share a unit. On the VT arm it is also at least the 95th percentile of within-culture SSU spread. SH and LSU carry no such floor, because their within-genome spread is wide and uneven (95th percentile: ITS 8.6%, LROR-FLR2 3.9%), and the merged units carry that variation instead.

2.6 LSU phylogenetic placement

The V18 references were realigned with MAFFT [15], and branch lengths and GTR+ Γ_4 parameters were fitted on the published topology with RAxML-NG [18]. Queries were added with `-addfragments -keeplength` and placed with EPA-ng [2]. The compatible edges of a query carry at least 95% of its likelihood weight. Edges inside a maximal clade bearing a single V18 species label are collapsed into one unit (the *species-clade* rule). Where a species label is not monophyletic on the backbone, its separate components stay separate units, and edges that carry several species labels stay individual units.

2.7 Reconciliation

A unit is linked to the species and genera of the training cultures whose culture set holds it. In each arm a copy is *matched* when its units belong to training cultures, *unmatched* when it has a call but no training culture holds the unit, and *silent* when it has no call. Genera are intersected over matched arms and species over the matched arms that name species. The call is a species, a species group, a genus, discordant or unplaced.

2.8 Novelty decision tree

The tree tests the cheapest explanation first.

1. **Artifact (A1).** Against the nearest main-type copy of another culture, the SSU differs by at least 1% and by at least half the ITS difference.
2. **Divergent class,** read from the copy itself. D1: the 5.8S is at least 3 bp shorter and at least 3% different from comparators of the baseline length. D2: the best-hit VT or SH is reached, in at least two other cultures, only by short-5.8S copies.
3. **Boundary (B1).** No VT call, but the best VT lies within one point of the 97% threshold.
4. **Novelty.** A species call is demoted to “novel species, genus known” when the VT or LSU arm is unmatched (N1, N2), or when the full-ITS distance to the nearest copy of any

training culture exceeds the 99th percentile of within-species distance between training cultures (N6). No matched arm gives “novel lineage”.

2.9 Species complexes

Some pairs of named species cannot be told apart from the markers. We define this from the reference alone. For each marker (full ITS, LROR–FLR2, SSU flank) the within-species spread is the 99th percentile of the distance from a main-type copy to the nearest main-type copy of another culture of the same species (ITS 6.0%, LSU 3.0%, SSU 0.8%). Two species are *inseparable* when, on every marker, their nearest copies lie no further apart than that spread: a copy of one then falls inside the spread of the other. Connected pairs form a complex, and a complex is reported as a unit wherever a species call is made (`species_groups.py`). The rule uses no test result and adds no threshold beyond the 99th percentile already used for the ITS rule, computed the same way (Python’s exclusive quantile method). To see how much the conclusions depend on that choice, the ITS rule and the complexes were also set together at the 95th and the 97.5th percentile: the rule was re-applied to the stored decision-tree rows with each fold’s limit refitted on its training cultures (`percentile_sensitivity.py`).

2.10 Validation design

Every culture-derived quantity is fitted on training cultures only within each fold: merged units, δ , unit-to-species links, D1 comparators, D2 support, the nearest-copy pools and the N6 threshold. *Culture holdout* has 148 folds, each removing one physical culture label together with all lineages sharing it. *Species holdout* has 50 folds, one per named species with all its cultures removed plus one per unnamed culture. A *false known* is a known species or species-group call for a copy whose species has no training culture.

In these benchmarks the MaarjAM, UNITE and V18 databases stay fixed. The external holdout (M4) removes each target species from V18 as well: V18 is realigned, the model refitted and all copies re-placed. MaarjAM and UNITE are not edited in any fold, so a withheld species keeps whatever virtual taxon and species hypothesis it has there; this matters because an unmatched virtual taxon demotes a call (N1) and an unmatched SH does not. Protocol v2 adds N6 and changes nothing else; *G. chinense* and *G. rugosae* informed that rule and are reported as development folds. The *first-outcome folds* are all other species folds: 34 in which the target’s V18 records were removed, and one (*Ambispora callosa*) whose species V18 does not hold by label, which is reported apart. The same protocol also has 15 genus folds, in which every culture of a genus and the genus’s V18 records are withheld together. The cohort was used to develop the methods, so all figures from it are development results; the independent test is described next.

2.11 Independent cultures

LongGloDB v1 (July 2026) [12] holds 1,122 curated SSU–ITS–LSU amplicon sequence variants (ASVs) from AMF culture collections. 810 of them carry the laboratory codes STFP and STFI and were left out: 784 of these are identical in full ITS to a copy in our reference, so they are the Stefani cultures reprocessed. 4 further cultures share a culture code with a Stefani label (3 of them under another species name) and were left out too (`longglodb_overlap.py`). The test set is the remaining 280 ASVs of 52 cultures from three other laboratories: 164 ASVs of 22 cultures of species the reference holds, 59 of species it does not hold in genera it does, 43 of genera it does not hold, and 14 named only to genus. The novel ASVs come from 26 cultures of 19 species.

Before any test sequence was searched, placed or scored, the complex rule (Section 2.9) was applied to the test ASVs, and its predictions were locked by hash together with a protocol and

two amendments that fix the outcomes, conditions and analyses (PROTOCOL_LGDB.md). A test ASV can be called a known species wrongly only if it lies within the within-species spread of that species on all three markers, so any false known outside the predicted set contradicts the result of Section 3.8. The frozen code of protocol v2 was used unchanged. All 151 reference cultures train and every test culture is excluded, so no test sequence enters merges, margins, unit links, comparators or the N6 threshold. Test sequences were cut by Stage 1 like the deposited copies, and their VT and SH evidence was computed with the same filtering against the same releases. In condition A the test windows are placed on the intact V18; in condition B each of the 19 novel species, all of which V18 holds by label, is removed from V18 in its own fold, as in M4.

The test amplicons begin about 60 bp inside the NS31–AM1 region, so a test SSU flank rarely covers 95% of a MaarjAM reference, and the frozen VT arm finds a passing hit for only 33 of 280 ASVs. The primary analysis applies the arm as it is. A secondary analysis, fixed before the run, measures VT coverage over the part of each reference the query spans and changes nothing else.

3 Results

3.1 Legacy windows on the deposited copies

All 904 deposited copies were anchored, and every window whose primers lie inside a copy was recovered (Table 2). Median lengths are close to the amplicons behind the legacy databases: 508 bp for NS31–AM1 with primers removed, about 550 bp with them, the size of the fragment behind MaarjAM virtual taxa. No window crosses an ITSxRust anchor, and no copy was flagged as a repeat. The 28 copies without an NS31–AM1 window are exactly the Paraglomerales copies, in which AM1 does not bind.

Table 2: Legacy barcode windows cut from the 904 deposited rDNA copies (amfsplit v0.3.1, primer-trimmed windows, up to 15% of each primer’s length in edits).

Window	Target	Usable	Missing	Open	Median (bp)
SSU NS31–AM1	MaarjAM VT	876	28 (3’)	0	508
SSU NS31–AML2	SSU studies	904	0	0	522
SSU WANDA–AML2	SSU V4	903	1 (5’)	0	500
SSU AMV4.5NF–AMDGR	SSU V4, short	904	0	0	216
LSU LROR–FLR2	Delavaux LSU	904	0	0	735
LSU FLR3–FLR4 (D2)	LSU D2	904	0	26 (3’)	330
AML1–LSUmBr	full amplicon	0	904 (both)	0	–

Usable = complete plus open-ended windows; missing = primer absent although the copy covers its position; open = site beyond the copy end. The 28 missing NS31–AM1 windows are all the Paraglomerales copies, in which AM1 does not bind; the one missing WANDA window is the copy dropped from the reference (PX215117.1). The 26 open D2 windows are copies that end before the FLR4 site. The outer amplicon primers are absent because the deposited copies are trimmed inside them. No window crosses an ITSxRust anchor, and none of the 904 copies is flagged as a concatemer.

3.2 Reference curation

Names are used as deposited, with genus synonyms (*Rhizoglossus* and *Rhizophagus*) treated as equal where genera are compared; the reorganisation of the Glomerales [6] postdates most deposits, and a name that is outdated in a database is not an error of the marker. The name check went against the expected relabelling. T3_IVT34A and CC4_4A_IVT7A are *R. irregularis* by ITS and LSU (for example ITS2 3.3% against 6.4% for T3), yet their SSU falls in VTX00113,

so VTX00113 holds both *R. irregularis* and *R. vesiculiferum*. In SD5_IVT32 the corrected main type matches V18 *R. proliferus* at 98.5–99.9%, while its short-5.8S copies are 91.8% from anything in V18. Two *Septoglomus* cultures, SO43BL and Blaszc_17_2277A, carry the same two lineages with the majority reversed; the lineages differ like species (ITS1 11–12%, 5.8S 5.7%, LSU 1.6%). Version 4 holds 903 copies in 151 cultures of 37 named species, with 51 divergent-class copies (43 short-5.8S, 8 same-length).

Table 3: Culture reference edits from the deposited copies to v4.

Item	Edit	Evidence
PX215117.1 (<i>G. rosea</i>)	dropped	SSU and ITS diverge at the same rate (4–6% and 5.6–7.6%); error or pseudogene
BR105_2435A	second lineage split off as <i>Paraglomus</i> sp.	three VTX00281 copies 28–33% from the culture’s <i>P. brasilianum</i> copies in ITS
4375_IVT88A, ON205A	labels merged	one culture filed under two organism names
T3_IVT34A,	kept as <i>R. irregularis</i>	name check: ITS2 3.3% vs 6.4% (T3); LSU nearest <i>R. irregularis</i>
CC4_4A_IVT7A		
SD5_IVT32,	main and divergent classes swapped	medoid fell in the short-5.8S group
212939_IVT38A		
SO43BL, Blaszc_17_2277A	mixed cultures split; SO43BL named <i>S. africanum</i>	reciprocal lineages; ITS and LSU agree on <i>S. africanum</i>

3.3 The three systems nest

The SSU flank assigns 895 of 903 copies to a VT. Counting cultures by their modal label, 3 of 37 named species span more than one VT and 8 of 30 VTs hold more than one species; VTX00199 alone holds four *Glomus* species. The LSU nearest reference gives a culture’s own species for 105 of 135 named cultures whose species V18 holds (78%; 24 get a congener and 6 no call) and the right genus for all that get a call, but V18 holds two records for nearly every species (460 records under 234 species labels: 215 labels with two records, five with three or four, 14 with one), and 45 of the 220 labels with two or more records hold two records under 97% apart, and in leave-one-out on V18 itself only 72% of records return their own species. The full UNITE+INSD set assigns 592 copies (66%) against 383 (42%) for the general release, which keeps one representative per SH. An SH is finer than a genome: the main-type copies of 74 of 107 cultures with two or more SH calls fall in two to 6 SHs. Counting each culture by its modal SH, 0 of 70 SHs hold two named species and VT splits 0 of 77 SHs; once SHs are merged into per-genome groups, one group is shared by *G. macrocarpum* and *G. rugosae*. 1 of 73 SHs is split by LSU (*D. varaderoana* / *D. bareai*) and 4 of 40 LSU units are split by VT. So SH mostly sits inside LSU species, which mostly sit inside VT.

3.4 Divergent copies and paralog units

The two divergent classes behave differently on every marker (Table 4). Short-5.8S copies are well represented in public data, consistent with the heterogeneous non-tandem units of a single genome [22]: their median best SH hit (99.8%) is closer than the main type’s (99.1%). Divergent copies reach 14 SHs; 11 of them no main-type copy reaches (paralog SHs, 4 under placeholder names), and 3 take one main-type copy each (Table 5). *S. viscosum*’s short copies are the only culture copies that reach VTX00409, about two points clear of VTX00063. In *Gigaspora rosea*, four cultures carry the same three or four rDNA types, identical across cultures and up to 6.2% apart in LSU, each in its own SH named *G. rosea*. These types are not confined to one isolate held in several collections: in LongGloDB (Section 3.9), INVAM cultures under other accession

codes (BR155B, FL105 and KS885; FL105 is INVAM’s reference culture for the species) carry ASVs within 0.25% of them in full ITS.

Table 4: The two divergent copy classes mislead each marker differently.

	Short 5.8S	Same length
Copies (cultures)	43 (24)	8 (5)
VT: same as the culture’s main type / other / no call	35 / 7 / 1	8 / 0 / 0
LSU: same species / another species / no call	5 / 1 / 31	2 / 5 / 1
LSU best-hit identity, median (main type 98.9%)	94.9%	97.2%
SH: same / other SH / no call	0 / 28 / 7	0 / 4 / 1
SH best-hit identity, median (main type 99.1%)	99.8%	98.6%

Each divergent copy against the modal unit of its culture’s main-type copies (crosswalk.py, culture reference v4). Copies in cultures without a main-type unit on a marker are left out of that marker’s row (short 5.8S: VT 0, LSU 6, SH 8; same length: VT 0, LSU 0, SH 3).

Table 5: Paralog units: SHs and a VT reached by divergent-class copies and by at most one main-type copy.

Unit	Name of the best hit	Divergent copies	Main	Identity
SH1337353.10FU	<i>Rhizophagus irregu-laris</i>	<i>R. irregularis</i> 10 (7 cultures)	1	100.0%
SH1364117.10FU	Glomeromycota sp.	[<i>R.</i>] <i>vesiculiferum</i> 7 (2 cultures)	0	98.5–99.8%
SH1081549.10FU	<i>Rhizophagus</i> sp.	<i>R. prolifer</i> 4 (1 culture)	0	99.3–99.8%
SH1081557.10FU	<i>Rhizogloinus intraradices</i>	<i>R. intraradices</i> 4 (3 cultures)	0	100.0%
SH1076762.10FU	<i>Gigaspora margarita</i>	<i>G. rosea</i> 2 (1 culture)	1	99.8%
SH1337340.10FU	<i>Rhizophagus</i> sp.	<i>Rhizophagus</i> sp. 1; <i>R. irregularis</i> 1 (2 cultures)	0	100.0%
SH1337458.10FU	<i>Rhizophagus irregu-laris</i>	<i>R. silesianus</i> 2 (1 culture)	0	100.0%
SH1389224.10FU	Glomeraceae sp.	<i>S. viscosum</i> 2 (1 culture)	0	100.0%
VTX00409	environmental (MaarjAM)	<i>S. viscosum</i> 3 (2 cultures)	0	98.5–99.4%

Main: main-type copies of any culture in the unit. 6 further units take one divergent copy each (SH1076712, SH1196937, SH1337247, SH1337463, SH1337480, SH1364178). Names are those of the best UNITE record; the VT row is MaarjAM’s.

3.5 Compatible sets

Within-culture SSU spread is small (median 0.07%, 95th percentile 0.54%, 99th 0.95%). Merging two VT pairs, VTX00113+VTX00115 and VTX00125+VTX00222, makes every culture consistent at $\delta = 0$ (Table 6). The VT spread floor ($\delta = 0.6$) costs three species of resolution. SH sets, once merged into per-genome groups, separate 32 of 37 species against 16 for VT; that figure is an upper bound because the merges are learned on the same cultures.

3.6 Reconciliation and novelty with the culture bridge held out

Combining the arms recovers what none reaches alone. Among the 755 main-type copies of species held by another culture, with merges and margins fitted on training cultures, the exact species comes from VT for 370 (49%), SH for 343 (45%) and LSU for 351 (46%), and from the combination for 607 (80%) before the decision tree; the tree demotes 4 of them, leaving 603 in Table 7. LSU placement raises exact-species calls and cuts false-known calls from 26.8% to 8.8%, and the ITS rule cuts them again to 2.2% (Table 7, Figure 1a); averaged over species or

Table 6: Calibrated compatible sets on culture reference v4.

Arm	Merged units	δ	Consistent cultures	One unit	Species resolved
VT	2	0.6	138/138	74%	16 of 37 (19 at $\delta = 0$)
SH	35	0	106/107	100%	32 of 37
LSU (nearest reference)	5	0	128/128	81%	27 of 37

A culture is consistent when the sets of its main-type copies share a unit. A named species is resolved when no other named species shares any of its unit groups. Merged units and species counts are learned and scored on the same cultures, so they are upper bounds.

over cultures the final rate is 1.4% and 1.9%. The ITS threshold was 6.0% in 139 of 148 culture folds. It demoted 6 known-species copies (4 from *G. chinense* 4036_906B, 1 from *G. rosea* CU131_1018E, 1 from *M. litorea* 5274); the *M. litorea* copy lies 12.6% from the species’ other culture. The 18 remaining errors are two complexes: *D. varaderana* with *D. aestuarii* (9 copies), which are indistinguishable on every marker, and *G. chinense* with *G. rugosae* (9 copies), whose ITS spreads overlap. Of the 42 divergent-class copies whose species is held by another culture, 35 were recognised as divergent class, 3 same-length copies were called their own species, and 4 were wrongly called a novel species. Short-5.8S copies were recognised from their own sequence in 36 of 43 cases with their culture held out and 29 with their species held out. Placement has a cost: 146 held-out copies lose a genus-level novelty call and become “novel lineage”.

Table 7: Training-only benchmark with the external databases fixed.

Configuration	Known kept (of 755)	Exact species	False known, all held out (of 803)	False known, single-culture (of 48)
Nearest-reference LSU	748	603	215 (26.8%)	3
LSU species-clade placement	751	701	71 (8.8%)	0
Placement + ITS rule (N6)	745	697	18 (2.2%)	0

Known kept: main-type copies of species held by another training culture that receive a known species or species-group call containing the true species (culture holdout, 148 folds). *False known*: known calls for copies whose species has no training culture (species holdout, 50 folds). Averaged over species and over cultures, the final false-known rate is 1.4% and 1.9%.

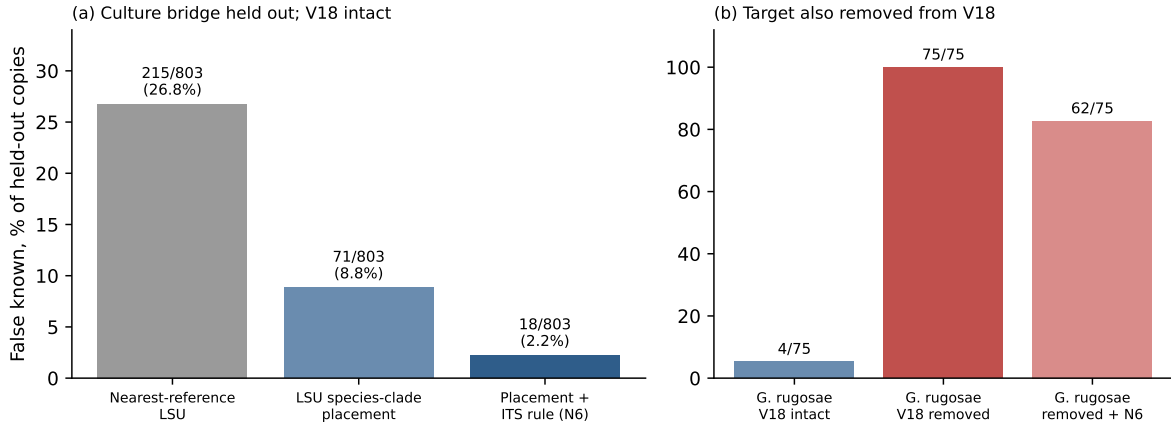


Figure 1: False-known calls. (a) Species holdout with the external databases fixed, by configuration. (b) *G. rugosae* with the species also removed from V18: the novelty signal that LSU placement gave came from V18 itself.

3.7 Removing the species from the external LSU reference

With the full V18, *G. rugosae* copies fall into V18's own *rugosae* clade, which no training culture holds, and are flagged as new. With the species removed they fall into the *chinense* and *macrocarpum* clades and are called known (Table 8, Figure 1b). The ITS rule does not use V18 but rescues only 13 of 75 copies, because *G. rugosae* ITS lies 1.4–7.5% (median 4.7%) from its relatives, inside ordinary within-species spread. Pooled over the 34 first-outcome folds, false-known calls among 673 copies were 19 with V18 intact, 78 with the target removed and 53 with the ITS rule (per fold in Table S1); averaged over species the rates are 20.6% and 11.7%. The 53 errors left after the ITS rule fall in 6 species pairs: *D. aestuarii* called *D. varaderana* (20), *R. irregularis* called *[R.] vesiculiferum* (18), *D. arenaria* called *D. jakucsiae* (4), *D. jakucsiae* called *D. arenaria* (4), *G. macrocarpum* called *G. rugosae* (4), *D. varaderana* called *D. aestuarii* (3). Of the 673 held-out copies, 272 were called a novel species in a known genus (195 because the LSU arm was unmatched, 34 because no species was shared by the arms, 25 by the ITS rule and 18 because the SSU arm was unmatched), 347 a novel lineage, which carries no genus, 53 a known species or group, and 1 was flagged as divergent class. The genus was called for 326 copies and was right for 326 of them, but that is precision: of the 487 copies whose genus is held by another culture, the genus was kept for 326 (66.9%), and 161 were called a novel lineage although their genus was known; the other 186 copies belong to genera with no other culture and were all called a novel lineage. The SSU arm's contribution to novelty is therefore small (2.7% of the copies), but it exists: it needs the withheld species to have a virtual taxon of its own. The one species V18 does not hold by label, *Ambispora callosa*, was called a known species for 20/20 copies without the ITS rule and 0/20 with it, because its ITS lies about 10% from its closest relative.

Table 8: Removing the target species from the LSU reference as well.

Target	V18	False known (v1)	With N6 (v2)
<i>G. chinense</i> [†]	full	28/35	5/35
<i>G. chinense</i> [†]	target removed	28/35	5/35
<i>G. rugosae</i> [†]	full	4/75	4/75
<i>G. rugosae</i> [†]	target removed	75/75	62/75
34 first-outcome folds	full	19/673 (2.8%)	9/673 (1.3%)
34 first-outcome folds	target removed	78/673 (11.6%)	53/673 (7.9%)

False-known calls among the main-type copies of each fold's target species.

[†] Development folds: these two species informed the N6 rule. *Full*: the fold's copies placed on the intact V18; *target removed*: V18 realigned without the target's records, model refitted, all copies re-placed. First-outcome folds pool the species whose V18 records were removed; 1 further fold(s) whose species V18 does not hold by label are reported apart.

With a whole genus withheld from the cultures and from V18 (15 folds, 803 main-type copies of named species), no copy was called a known species, with or without the ITS rule (0 and 0); this is expected, because no training culture shares the genus. 734 copies were called a novel lineage, the right rank. The other 69 (8.6%) were called a novel species in a known genus, and that genus was wrong in every case: *Rhizophagus* called *Sclerocystis* (69). With V18 intact, 0 copies received a wrong genus. The genus fold shows at the level of genera what the species folds show at the level of species: a copy whose clade has been removed is placed on the sister clade, here *Sclerocystis*, and named by the training cultures that hold it. None of the 51 divergent-class copies was recognised as divergent in these folds (0), because D1 and D2 both need a related culture, and 40 of them also received the wrong genus.

3.8 The remaining errors are species complexes

4 complexes emerge from the distance rule (Table 9): *G. chinense*, *G. macrocarpum* and *G. rugosae* (the VTX00199 group); *D. aestuarii* with *D. varaderana*; *D. arenaria* with *D. jakucsiae*; and *R. irregularis* with *[R.] vesiculiferum*. Together they hold 9 of 37 named species and 29.9% of the reference copies. One further pair, *D. celata* with *D. eburnea*, misses the LSU limit by 0.4 points and is not a complex. The rule depends on the percentile. *D. aestuarii* with *D. varaderana*, *D. arenaria* with *D. jakucsiae*, and *G. macrocarpum* with *G. rugosae* are complexes at every percentile from the 95th to the 99.5th; *G. chinense* joins from the 97.5th, *R. irregularis* with *[R.] vesiculiferum* from the 99th, and *F. coronatus* with *F. mosseae* at the 99.5th.

The ITS rule uses the same percentile, so a fair test moves both together (Table 10). At the 97.5th percentile the rule leaves the same 53 false-known calls as at the 99th, but the 18 copies of *R. irregularis* called *[R.] vesiculiferum* now lie outside a complex. At the 95th the rule is stricter: false known falls to 37 (5.5%), 5 of them outside a complex, all again *R. irregularis* called *[R.] vesiculiferum*, at the cost of demoting 32 known-species copies in the culture holdout (4.2%) instead of 6. At every percentile, then, each remaining error is either inside a complex or this one pair, and whether the pair counts as a complex is the only conclusion that depends on the choice.

Every one of the 53 false-known calls left after the ITS rule in the first-outcome folds is a call to a member of the species' own complex. Without the ITS rule, 54 of the 78 errors are inside a complex and 24 (3.6%) are not; with V18 intact the corresponding figures are 10 and 0 copies outside a complex. So the ITS rule removes the errors that the markers could, in principle, have avoided, and what is left is the limit of the markers themselves. The complexes were derived from all cultures, including the withheld species, so this shows that the errors are confined to pairs the reference already shows to be inseparable; it does not show that an undescribed relative of a complex member would be caught, which it would not, by definition.

Table 9: Species complexes: pairs of named species whose nearest copies lie within the within-species spread on all three markers.

Species A	Species B	Complex	ITS	LSU	SSU
<i>Diversispora aestuarii</i>	<i>Diversispora varaderana</i>	C2	1.1	1.09	0.0
<i>Diversispora arenaria</i>	<i>Diversispora jakucsiae</i>	C3	3.48	0.83	0.07
<i>Glomus chinense</i>	<i>Glomus macrocarpum</i>	C1	5.87	2.18	0.27
<i>Glomus chinense</i>	<i>Glomus rugosae</i>	C1	4.95	1.36	0.2
<i>Glomus macrocarpum</i>	<i>Glomus rugosae</i>	C1	1.38	1.36	0.07
<i>Rhizophagus irregularis</i>	<i>[Rhizoglomus] vesiculiferum</i>	C4	3.25	2.44	0.34

Columns give the distance (%) between the nearest copies of the two species. The limits are the 99th percentile of the distance between cultures of one species: ITS 6.0%, LSU 3.0%, SSU 0.8%.

Table 10: The ITS rule and the species complexes moved to the same percentile.

Percentile	ITS limit (median)	Complexes (species)	Known demoted (of 755)	False known, species holdout (of 803)	False known, V18 removed (of 673)	Outside a complex
95th	4.4%	3 (6)	32 (4.2%)	9 (1.1%)	37 (5.5%)	5
97.5th	5.0%	3 (7)	22 (2.9%)	13 (1.6%)	53 (7.9%)	18
99th	6.0%	4 (9)	6 (0.8%)	18 (2.2%)	53 (7.9%)	0

The ITS rule is re-applied to the stored decision-tree rows at each percentile, with the limit fitted on each fold's training cultures, and the complexes are those of the distance rule at the same percentile (`percentile_sensitivity.py`). *Known demoted*: known-species copies the rule demotes in the culture holdout. *V18 removed*: first-outcome M4 folds. The 99th-percentile row reproduces the reported results exactly. Every error outside a complex is *R. irregularis* called *[R.] vesiculiferum*.

3.9 Cultures from other collections

Before any test result was scored, the 850 copies of the 37 M4 species folds were rescored with the test cultures present and excluded, and every call came back as stored. Adding the test windows to the V18 alignment left the 903 reference alignments and credible unit sets unchanged.

Among the 102 ASVs of species the reference does not hold, 4 were called a known species with the target removed from V18 (3.9%; 5.3% averaged over species, 3.8% over cultures; Table 11). They are the 4 ASVs the locked prediction placed inside a reference species, *Gigaspora gigantea* KS302 called *G. rosea*, and 0 lie outside the prediction. With V18 intact their LSU falls in V18’s own *Gigaspora gigantea* clade, which the reference’s *G. rosea* cultures also occupy, and their ITS lies 0.5–2.0% from those cultures, so no marker separates the two. None of the 43 ASVs from genera the reference does not hold was called known. Removing the species from V18 changed no call: 102 of 102 novel ASVs had the same status and placement in conditions A and B, because their LSU fell, in both, in clades no training culture holds. These species are far from the reference’s, and the sister-species case behind the external-removal result (Section 3.7) occurred once, in KS302, where it was predicted.

Of the 164 ASVs of species the reference holds, 136 kept their species (82.9%), against 745 of 755 in the culture holdout. ASVs identical in ITS to a reference copy kept it in 40 of 43 cases, the others in 96 of 121 (79.3%). The ITS rule demoted 15 of them, whose ITS lies 6.1–9.6% from every reference culture, beyond its 6.0% threshold: without the rule 151 kept their species (92.1%), and it removed no false-known call (4 without it). The 8 ASVs called another species are *Funnelformis coronatus* MR101 called *F. mosseae*, which the lock had predicted (8 of 8): their ITS lies 0.0–0.6% and their SSU 0.0% from reference *F. mosseae* cultures, so the culture is most likely *F. mosseae* under another name. The lock sorted the known species: 135 of the 139 ASVs predicted inside their own species kept it, and 24 of the 25 predicted outside did not.

With the frozen VT arm nearly silent, the genus was hard to name: 44 of the 59 ASVs of novel species in genera the reference holds were called a novel lineage and 9 a VT boundary, and the genus was given for 6, all right. With VT coverage measured over the spanned part, the genus was given for 29, again all right, but two errors appeared: the 18 ASVs of *Racocetra gregaria* were given *Cetraspora*, through a VT the two genera share (VTX00041) while their LSU was unmatched; and one ASV, *Septoglomus constrictum* KS890 called *S. africanum* / *S. turnauae*, through a shared VT, its ITS 1.1% from a reference culture. False-known calls then number 5 (4.9%), 1 of them outside the lock.

The test also raised curation questions for the collections. Three culture codes carry different species in LongGloDB and in the Stefani deposit (MT106, ON205A and BEG9). INVAM’s name for both MT106 and ON205A is *A. gerdemannii*, which Stefani et al. record and depart from (their Table S2) and LongGloDB keeps for ON205A. In addition, the LongGloDB *A. leptoticha* MX982A lies 7.8–9.6% in ITS from the reference’s *A. leptoticha*, ON205A, which LongGloDB names *A. gerdemannii*.

4 Discussion

Linked copies turn the VT–SH–LSU correspondence from a guess into a measurement, and show that each naming system fails in its own predictable way.

The bridge is mostly nested, not one-to-one. SHs mostly sit inside LSU species, which mostly sit inside VTs. VT counts therefore underestimate species (at best a VT resolves 19 of 37), and SH counts overestimate them, because one genome spans up to six SHs. Surveys that count SHs as taxa inflate AMF richness, and part of that inflation is intragenomic. Grouping SHs per genome restores most of the species signal in our cultures, but the groups are learned from these same cultures and their transfer to field sequences is untested. The 2.5 kb amplicons were

Table 11: Independent cultures: the frozen rules on 52 LongGloDB cultures from other collections.

Analysis	Condition	False known (of 102)	Share	Novel genus wrong / 59	species: right / none (of 43)	Novel genera: given (of 43)	Known kept (of 164)
Primary	A: V18 intact	4 (4)	3.9%	6 / 0 / 53	0		136 (151)
	B: target removed	4 (4)	3.9%	6 / 0 / 53	0		–
VT spanned	A: V18 intact	5 (5)	4.9%	29 / 0 / 30	18		139 (154)
	B: target removed	5 (5)	4.9%	29 / 0 / 30	18		–

Protocol v2 (with the ITS rule, N6); in brackets, the same without it. *False known*: ASVs of species the reference does not hold called a known species or species group. *Novel species*: ASVs of species the reference does not hold, in genera it holds. *Novel genera*: ASVs of genera it does not hold, so any genus given is wrong. *Known kept*: ASVs of species the reference holds whose call contains their species; condition A only. *VT spanned*: the secondary analysis, with VT coverage measured over the part of each MaarjAM reference the query spans (amendment 2).

introduced to consolidate the three regions into one dataset [16]; what is missing is a measured correspondence between the names, which the linked copies supply.

Divergent copies are the main source of false taxa. Intragenomic rDNA variation in AMF is long established [28, 34], yet it has been largely overlooked in ecological and taxonomic work [32]; what the copies add is a measurement of how it misleads each marker. Short-5.8S copies look like novel lineages in LSU and form their own SHs and VTs, several carrying placeholder names in UNITE. The D1 test recognises most of them from sequence alone, but recognition falls when the whole species is unknown, since it then lacks a related culture of the right 5.8S length to compare with. Same-length divergent copies are more ambiguous: co-dominant types of one genome in *Gigaspora*, second lineages in *Septoglomus*. They should be flagged, not explained away.

Novelty has a floor set by the markers. 4 complexes, 9 species in all, cannot be separated on SSU, ITS and LSU together: their nearest copies lie inside each other’s within-species spread on every marker. All errors that remain after the ITS rule fall inside them at the 99th percentile used throughout. The claim holds for every percentile we tried except for one pair: with the rule and the complexes both at the 95th or 97.5th percentile, *R. irregularis* and *[R.] vesiculiferum* are no longer a complex, and the *R. irregularis* copies called *[R.] vesiculiferum* are errors outside one (Table 10). Whether those two names are separable on rDNA is therefore the open question, not whether the floor exists. A species call there is honest only as a group, and a novelty call there is not possible: an undescribed relative inside the spread of a known species is indistinguishable from that species on these regions. The alternative reading is that some of these names are not separate lineages, a question for taxonomists; *D. aestuarii* and *D. varaderana* differ by 0.0% in SSU and 1.1% in LSU. The rule is data-driven and will grow as cultures are added: one further pair (*D. celata* with *D. eburnea*) misses the LSU limit by 0.4 points.

Novelty depends on what the reference already knows. With only the culture bridge held out, LSU placement looks like a strong novelty detector, but much of that strength comes from the withheld species still being in V18: its clade holds no culture, so the copy is flagged. Phylogenetic placement was proposed because database matching cannot place sequences from taxa that are absent from the database [9, 7]. That holds at the level of clades, but our external-removal folds show its limit at the level of species: with the species removed, a copy is placed on its nearest named relative and then called by it, and with a genus removed, on the sister

genus (69 *Rhizophagus* copies called *Sclerocystis*). An undescribed species is absent from every reference. Removing it from V18 is therefore closer to the real situation than leaving it in, but not equal to it: MaarjAM and UNITE were left intact, and many virtual taxa are built from environmental sequences, so a species that is undescribed yet already sequenced in the field can still have a virtual taxon of its own. The SSU arm’s novelty signal accounts for 18 of the 673 copies, so for a species never sequenced the false-known rate may be higher than reported by up to about 2.7% of the copies. The culture-ITS rule does not depend on external databases and repairs part of the gap, but not where ITS itself overlaps. Pooled over the first-outcome folds, false known rose from 2.8% with V18 intact to 11.6% with the target removed, and fell to 7.9% with the ITS rule.

Relation to earlier work on the same cultures. Stefani et al. [32] sequenced the copies used here (148 cultures of 44 species) and compared rDNA with three protein-coding genes (glomalin, H⁺-ATPase and RPB1), sequenced for 143 cultures of 39 species. They found no barcode gap in any of the regions, so no single distance threshold identifies AMF, and concluded that rDNA is still the practical choice because its primers work across all lineages. Our results agree where they overlap: the complexes are pairs whose nearest copies lie inside each other’s spread on every rDNA region. The questions differ. They ask whether a distance threshold identifies a species within one marker; we ask how the three naming systems relate and how much novelty can be called once the intragenomic classes are accounted for. Their protein-coding sequences are public (glomalin PV808915–PV809055, RPB1 PV809056–PV809196, H⁺-ATPase PV809197–PV809333) and could settle the complexes: if those genes separate *D. aestuarii* from *D. varaderana*, the complex is a limit of rDNA, and if not it is a question of naming. We compared them (`complex_pcgenes.py`; Table S2). At the thresholds Stefani et al. proposed, the three genes separate almost every congeneric pair that rDNA separates (42 of 44 on glomalin; 43 of 45 on RPB1; 43 of 45 on H⁺-ATPase; 95–96%). Of the 6 complex pairs, 4 have protein-coding records for both species. Of these, 2 exceed the threshold on one gene each (*D. aestuarii* and *D. varaderana* on RPB1 (1.24% against 1.1%); *R. irregularis* and [*R.*] *vesiculiferum* on glomalin (1.32% against 1.0%)), but none of the 12 comparisons exceeds the spread within the species themselves. The complexes are therefore not a limit of rDNA alone: on these genes, too, their members lie as close as cultures of one species, and whether they are separate species is a question for taxonomy rather than for the choice of marker. Joining the records through culture names also showed that the RPB1 and H⁺-ATPase deposits rotate the species names of three cultures: FS16_2427A (*Diversispora epigaea* in the reference, *Septoglomus* sp. in GenBank); FS_9_2328B (*Septoglomus* sp. in the reference, [*Rhizoglomus*] *vesiculiferum* in GenBank); Gwalker_i_IVT132A ([*Rhizoglomus*] *vesiculiferum* in the reference, *Diversispora epigaea* in GenBank). Each of these records lies 0.0–0.9% from records of its culture’s species, so the sequences are right and the names are not; we left the 6 records out.

Why the rules are not calibrated per species. A calibrated novelty flag exists for fungal ITS generally [25]. Here the culture-ITS threshold is itself a calibrated quantile, the 99th percentile of the distance between cultures of one species, and it behaves as set: it demoted 6 of 755 known-species copies (0.8%). The 99th percentile was fixed in protocol v2 before the first-outcome folds were run; the 95th would have cut false-known calls further (5.5% against 7.9% with the species removed from V18) at the cost of demoting 4.2% of known-species copies instead of 0.8% (Table 10), a trade-off a user can set. The other rules, such as the unmatched-arm demotions, are not calibrated, because the reference holds too few cultures per species to fit a threshold for each, and because the AMF-specific traps (divergent classes, species complexes) have to be handled before a calibration can mean anything.

Independent cultures. On 52 cultures from three other laboratories, with the predictions locked before the run, the frozen rules called a species the reference does not hold a known species only where the complex rule said they would: *G. gigantea* KS302, inside *G. rosea* on every marker. The test is easier than ours in one respect and harder in two. Its novel species are far from the reference’s, so it says little about the sister-species case, which occurred once. Its amplicons start inside the VT region, which leaves the VT arm silent unless coverage is counted over the part a query spans, and a genus taken from VT alone was then wrong when the true genus was absent from the reference (*Racocetra* given *Cetrastoma*). And its cultures come from other collections, some of which lie further from ours in ITS than our cultures of one species lie from each other: the ITS rule, fitted at the 99th percentile on one collection, demoted 15 of 164 known-species ASVs and prevented no false known. Its threshold should be fitted on more than one collection before the rule is used on field data.

Implications for surveys. Name at the level the markers can support: report VT sets, SH groups per genome and LSU species clades, with species groups where the markers cannot decide. Screen for the short-5.8S class before counting taxa, and treat paralog SHs and VTs, such as SH1337353 and VTX00409, as known genomes. Rest novelty claims on evidence that does not come from the database that defines “known”, and read names against a taxonomy that is stated, because outdated names in a database look like marker errors [6]. Replication across independent libraries is planned for environmental data.

4.1 Limitations

Apart from the LongGloDB test, all results come from one collection of cultured species, and the cohort was used to develop the methods, so they are development estimates. The LongGloDB test rests on 26 novel cultures, covers distant novelty, and used amplicons on which the frozen VT arm is silent. The 99th percentile behind the ITS rule and the complexes was chosen after seeing in-sample distances (Table 10 shows the effect of other choices), and the complexes are derived from all cultures, including the withheld species. Culture names are treated as truth, though some are doubtful: *M. litorea* 5274, some *G. rosea* cultures, and *Septoglomus* sp. C4 and 5070_5024A, whose LSU matches *Microviscospora peruvioscosa*. MaarjAM and UNITE stay intact in every benchmark; the external holdout removes species from the LSU reference only. About a quarter of V18’s multi-record species labels (51 in the placement audit) do not form exclusive clades on its backbone. Independent-library replication on environmental data and the Stage 2 abundance model are outstanding.

5 Conclusions

A novelty call for an AMF sequence is only as good as the external references are complete, and it cannot be made below the resolution of the markers: with the target species removed from the LSU reference, a share of false known calls remains, and every one of them lies in a species complex that SSU, ITS and LSU cannot separate. On cultures from other collections, the only species wrongly called known was the one the complex rule predicted. Linked long-read copies give an empirical bridge between the three naming systems and show where each one misleads. Divergent copies must be recognised before taxa are counted, and species that the markers cannot separate, which protein-coding genes do not separate either, should be reported as groups.

Data and code availability

Sequences are GenBank PX207096–PX207274 and PX214440–PX215164. The culture reference (version 4, 181938d695f3df40), the edits files and their manifests, the Stage 3 scripts and the validation packages (training-only bridge, LSU placement, external holdout with protocols v1 and v2) will be deposited with the paper, as will the LongGloDB test package (protocol, amendments, locked predictions, runner and score tables); LongGloDB v1 is available from GloDB [12]. [\[TODO: repository URL and archive DOI\]](#) External references: MaarjAM gDAT snapshot (sha256 2b2c57e664e76347...), UNITE 10.0 of 19 February 2025 [1] (UNITE+INSD sha256 abe98f43e16c179f...) and Delavaux V18 AMF-only (sha256 e6794cbb166735a...).

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Author contributions

Aaron O’Brien conceived the study, developed the software, carried out the analyses and wrote the manuscript.

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Competing interests

The author develops ITSxRust [24], which is used in this study, and declares no other competing interests.

Use of generative AI

Anthropic’s Claude and Microsoft Copilot (the versions current in 2026) were used in preparing this work: to draft and revise manuscript text and to write and debug analysis scripts. The author revised the drafted text extensively, ran all analyses, verified all reported values against the primary output, and is responsible for the content of the manuscript.

A Supplementary tables

Table S1: External holdout by species fold: false-known calls among the target species’ main-type copies.

Target species	Copies	Intact	Intact + N6	Removed	Removed + N6
<i>Acaulospora morrowiae</i>	10	0	0	0	0
<i>Ambispora callosa</i> [‡]	20	20	0	20	0
<i>Ambispora leptoticha</i>	10	10	0	10	0

Target species	Copies	Intact	Intact + N6	Removed	Removed + N6
<i>Cetraspora gilmorei</i>	60	0	0	0	0
<i>Dentiscutata heterogama</i>	6	0	0	0	0
<i>Diversispora aestuarii</i>	20	1	1	20	20
<i>Diversispora arenaria</i>	4	0	0	4	4
<i>Diversispora celata</i>	10	0	0	0	0
<i>Diversispora eburnea</i>	15	0	0	0	0
<i>Diversispora epigaea</i>	13	0	0	0	0
<i>Diversispora jakucsiae</i>	4	0	0	4	4
<i>Diversispora spurca</i>	18	0	0	0	0
<i>Diversispora varaderana</i>	31	8	8	3	3
<i>Dominikia bonfanteae</i>	6	0	0	0	0
<i>Dominikia duoreactiva</i>	9	0	0	0	0
<i>Funneliformis caledonium</i>	9	0	0	0	0
<i>Funneliformis coronatus</i>	24	0	0	0	0
<i>Funneliformis geosporum</i>	8	0	0	8	0
<i>Funneliformis mosseae</i>	130	0	0	0	0
<i>Gigaspora rosea</i>	51	0	0	0	0
<i>Glomus chinense</i> [†]	35	28	5	28	5
<i>Glomus macrocarpum</i>	7	0	0	4	4
<i>Glomus rugosae</i> [†]	75	4	4	75	62
<i>Glomus tetrastratosum</i>	21	0	0	0	0
<i>Microdominikia litorea</i>	12	0	0	0	0
<i>Oehlia diaphana</i>	29	0	0	0	0
<i>Paraglomus brasilianum</i>	8	0	0	0	0
<i>Paraglomus laccatum</i>	5	0	0	0	0
<i>Rhizophagus intraradices</i>	12	0	0	0	0
<i>Rhizophagus irregularis</i>	56	0	0	19	18
<i>Rhizophagus prolifer</i>	6	0	0	0	0
<i>Rhizophagus silesianus</i>	7	0	0	0	0
<i>Sclerocystis sinuosa</i>	18	0	0	0	0
<i>Septoglomus africanum</i>	6	0	0	6	0
<i>Septoglomus turnauae</i>	4	0	0	0	0
<i>Septoglomus viscosum</i>	21	0	0	0	0
<i>[Rhizoglomus] vesiculiferum</i>	23	0	0	0	0

Columns: false-known calls with V18 intact, V18 intact plus the ITS rule (N6), the target removed from V18, and removed plus N6. [†] development fold; [‡] species not in V18 by label (removal changes nothing).

Table S2: Protein-coding genes on the rDNA species complexes: smallest distance (%) between the two species, with the largest distance within either species in brackets.

Pair	Glomalin	RPB1	H ⁺ -ATPase
<i>D. aestuarii</i> / <i>D. varaderana</i>	0.73 (0.0)	1.24 (1.73)	1.22 (0.0)
<i>D. arenaria</i> / <i>D. jakucsiae</i>	0.29 (–)	0.57 (–)	1.55 (–)
<i>G. chinense</i> / <i>G. rugosae</i>	0.73 (0.73)	0.33 (7.93)	1.00 (1.32)
<i>R. irregularis</i> / <i>[R.] vesiculiferum</i>	1.32 (1.61)	0.81 (7.54)	1.18 (5.77)

Sequences of Stefani et al. (2025). Indicative species thresholds from that study: 1.0% / 1.1% / 1.7% (glomalin / RPB1 / H⁺-ATPase). Distances are edlib infix edit distances including any intron, so they are comparable with each other but only approximately with those thresholds. Records are joined to species through the reference culture names; records whose GenBank name disagrees with the reference name of their culture are left out (6).

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